

1. Two balls are randomly drawn from a bowl containing 6 white and 5 black balls. What is the probability that one of the drawn ball is white and the other is black?

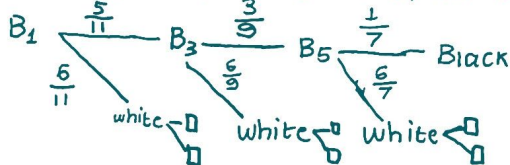
$$P = \frac{\binom{6}{1} \binom{5}{1}}{\binom{11}{2}} = \frac{6 \cdot 5}{66} = \frac{6}{11} = 54,55\%$$

2. A student committee of size 5 is to be randomly selected from the group of 6 men and 9 women. Calculate the probability that the committee consist of 3 men and 2 women

$$P = \frac{\binom{6}{3} \binom{9}{2}}{\binom{15}{5}} = \frac{20 \cdot 36}{3003} = 23,976\%$$

3. A basketball team consist of 6 black and 6 white player. The players are to be paired in groups of two for the purpose of determining roommates. If the pairing are done at random, what is the probability that none of the black player will have a white roommate?

$\{B_1, B_2, \dots, B_6\}$ and $\{W_1, W_2, \dots, W_6\}$



$$P = \frac{5}{11} \cdot \frac{3}{9} \cdot \frac{1}{7} = \frac{5}{231} = 2,16\%$$

4. If n people are present in a room, what is the probability that no two of them celebrate birthday on the same day of the year? How large need n be so that this probability is less than 0.5 (50%)?

$$P(n) = \frac{365 - n + 1}{365} \cdot P_{n-1}$$

$$P(22) = 0,5243 = 52,43\%$$

$$P(23) = 0,4927 = 49,27\%$$

$$P(n) = \frac{365!}{365^n \cdot (365 - n)!}$$

$$= \frac{365 P_n}{365^n} = \frac{P(365, n)}{365^n}$$

at least $n=23$ so that the probabability is less than 0.5